Ryan Ruben Pascual

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Education

Rice University Houston, TX

Bachelor of Arts in Computer Science *Expected May 2028*

- **GPA**: 4.0/4.0
- **Awards**: USACO Platinum, Codeforces Master

Experience

Software Developer Intern (Summer, continued part-time) May 2026 – October 2026

SAS Institute *Cary, NC*

- Re-architected a Go ETL pipeline (Azure Blob / FHIR API → Snowflake) into concurrent download, transform, and load stages to be **6x faster**; now ingests **10+ TB monthly** in production, saving 300+ hours of monthly ingestion runtime
- Optimized an internal Go medical data transformation package used by the entire Health & Life Sciences Division, reducing medical FHIR transformation latency by **90%** through parallel parsing and caching
- Built an open-source Python MCP server used by **40+ internal employees and 15+ clients**, enabling LLM agents to automate SAS Health data ingestion, ML pipeline construction, schema mapping, and connection management through OAuth-authenticated SAS Viya APIs
- Designed an async job-execution layer for the MCP server so agents can run 2+ hour Viya ML jobs without blocking, collapsing a 50-step manual SAS Model Studio workflow into one prompt

Technical Lead May 2026 – Present

RiceApps, Rice University *Houston, TX*

- Architected and led development of an OpenAI-compatible LLM inference gateway (Go, PostgreSQL, Redis) serving 15+ projects, centralizing access and cost tracking to **cut club-wide AI spending by 30%**
- Implemented capability-aware multi-provider routing with per-model circuit breakers and automatic failover, so client projects keep serving requests when an upstream provider goes down
- Mentored **10 engineers** and guided new developers from onboarding through implementation, leading weekly architecture and code reviews

Software Engineer September 2025 – May 2026

RiceApps, Rice University *Houston, TX*

- Delivered an end-to-end Next.js/TypeScript + PostgreSQL inventory system for Fondren Library tracking 200+ assets, saving **2,600 paper forms** and **87 hours** of staff time annually

Projects

hopper, Distributed Job Queue | *Go, gRPC, PostgreSQL, Kubernetes* GitHub

- Built a fault-tolerant job queue that keeps background work running through server crashes, processing **1M+ jobs with zero lost** while workers, servers, and the database were deliberately killed mid-run
- Increased throughput from **2K to 9K+ jobs/sec** by batching database writes and switching from polling to push-based notifications in PostgreSQL
- Scaled workers automatically from 1 to 20 based on queue backlog using Kubernetes and KEDA, so capacity grows with demand without manual intervention

SkyTrack, Flight Delay Predictor | *Java, Spring Boot, AWS (SQS, DynamoDB, Lambda, S3)* GitHub

- Built a cascade delay model that propagates a late arrival through each aircraft's tail rotation across multiple legs, predicting downstream delays within **8.1 min mean error** and **88% precision** across 1,910 flights
- Ingested **1M+ live position updates/hour** across **7K aircraft** through SQS FIFO, partitioning by aircraft to preserve per-aircraft ordering while consumers process in parallel
- Designed a single-table DynamoDB schema for low-latency reads and archived history to partitioned S3 Parquet; added Prometheus/Grafana alerting and **400+ tests** in GitHub Actions CI against a Dockerized AWS environment

Technical Skills

Languages: Go, Java, Python, TypeScript, SQL
Backend & Data: gRPC, REST, Spring Boot, PostgreSQL, Redis, DynamoDB, Snowflake
Infrastructure: Kubernetes, Docker, AWS, Azure, Prometheus, GitHub Actions